



MINING DEALER BEST PRACTICE

Large Hex Stick Cylinder Transport Brackets

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

When the large mining excavators are being assembled and disassembled during build-up and dismantled transportation, the stick cylinders are generally left to hang loose by the barrel end being pinned to the boom. They tend to rest on the hydraulic and grease lines. This can create a situation for damage to the hydraulic and grease lines on the top of the boom when bounced around. The transport brackets described in this Best Practice helps in preventing the cylinders from hitting the lines and thus avoid damage to the hydraulic lines and grease lines.

2.0 Best Practice Description

During assembly of the new machine, the boom and stick cylinders are generally attached to the boom first before the whole unit is lifted up to be attached to the upper structure. This reduces the working height if the cylinders were to be fitted after the boom alone was fitted first. The boom has lugs welded on the side for attaching the boom cylinders. The stick cylinders do not have any securing mechanisms. Figure 1 shows the boom assembled and ready for attaching to the upper structure. The boom is also removed in this manner when certain repairs are performed. Such repairs would include boom pin/bush replacements and boom repairs. The boom can also be removed in this manner for transportation due to weight restrictions. Figure 2 shows the top surface of the boom where hydraulic lines and grease lines are subject to crushing potential if the stick cylinders are not restrained.



Fig 1: Boom with assembled boom and stick cylinders and ready for attaching to upper structure.



Fig 2: Potential crushing incidents of hydraulic and grease lines by unrestrained stick cylinders.

A dealer had made brackets to restrain the stick cylinders after installing into the boom. The stick cylinders are then ready for connection to the stick after installation of the boom but will not be loose during assembly. Figure 3 shows a clearer view of the bracket in use on the left hand side of the boom. The brackets rest on the boom by the cylinder weight and the cylinders are restrained from sideways movements by the cylinder pins at the barrel ends.

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Fig 3: Bracket in use

Stick Cylinders		
Hex Model	Approx Weight (kg)	Approx OD (mm)
6060	3900	420
6040	3000	390
6030	3000	370

Fig 4: Tabulated stick information.

Figure 4 shows a table of approximate weights and outside diameter of the stick cylinders for three different models. These figures are for reference when fabricating the brackets. Figure 5 is a drawing with indicative dimensions for a 6060 Hex.

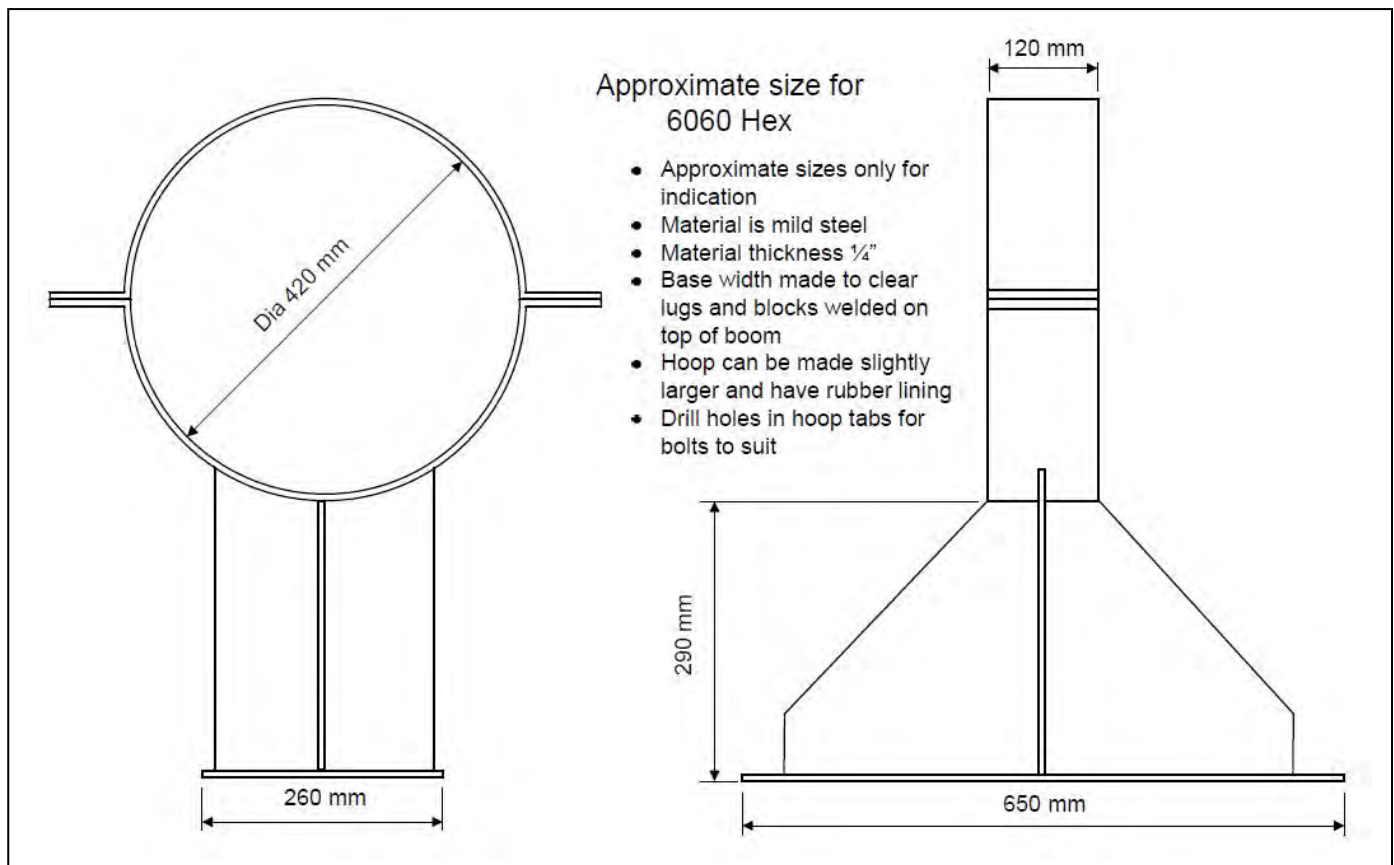


Fig 5: General drawing for 6060 Hex bracket as an indication

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3.0 Implementation Steps

1. Review delivery orders and available machines in territory or site.
2. Determine models required.
3. Investigate local legislations and regulations for supporting tools. Follow requirements if any.
4. Determine final dimensions from existing machines on site.
5. If there are no existing machines, make final dimensions when components arrive.
6. Fabrication brackets.

4.0 Benefits

1. Safety with more secure stick cylinders.
2. Cost savings in parts and labor by preventing accidental damage to hydraulic and grease lines, and mounting brackets on the top of boom.
3. Applies anytime the boom is removed or installed during new machine assembly, or during repairs.

5.0 Resources Required

Labor is required to investigate, obtain final measurements and fabricate.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

WesTrac Equipment
Western Australia
Australia

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